



Jurusan Teknologi Informasi Politeknik Negeri Malang
Mata Kuliah Data Warehouse
Kuis 1

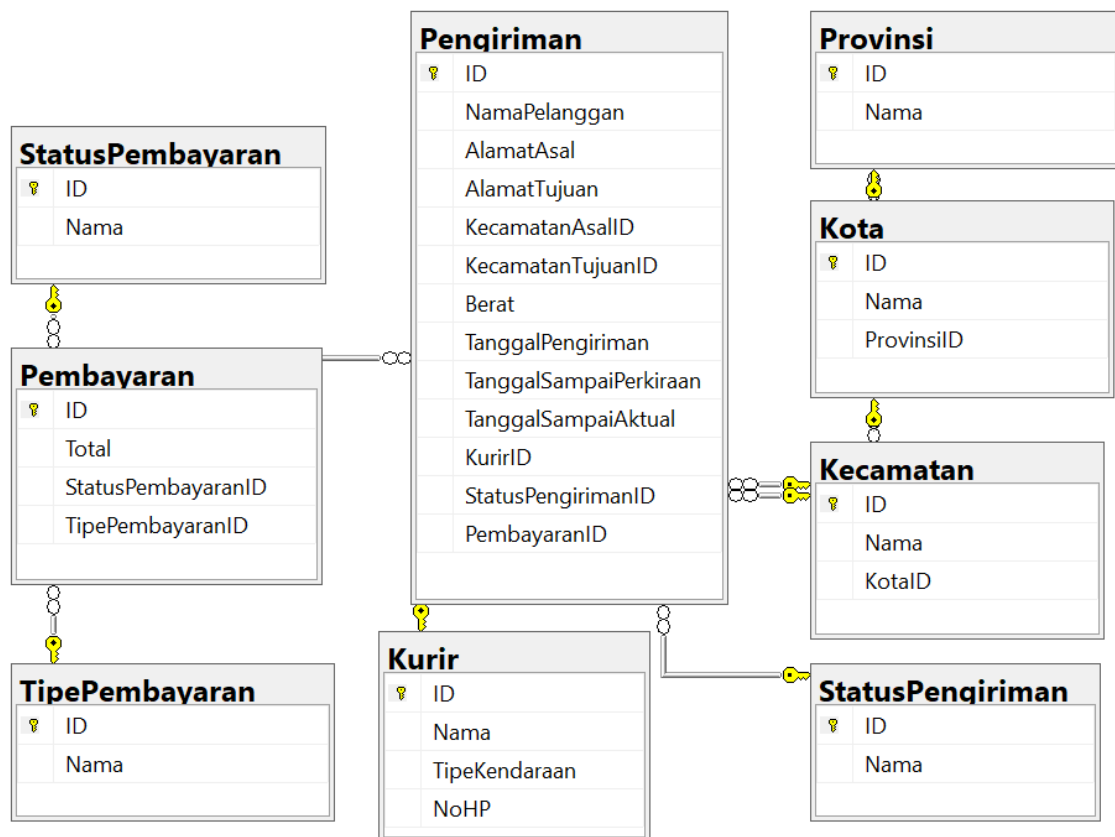
Nama : Hanin Mariam Abiyyah Hendrik

Nomor Urut : 12

1. Tuliskan perbandingan star schema dan snowflake schema pada tabel berikut:

	Star Schema	Snowflake Schema
Normalisasi	Data is not normalized (contains duplicated data)	Data is normalized (minimizes data duplication)
Kompleksitas desain/skema	Simpler design with fewer tables	More complex design with more relational tables
Kompleksitas query	Queries are simpler as they involve fewer tables	Queries are more complex due to multiple table joins
Performa query	Faster performance because fewer joins are required	Can be slower as more joins are needed
Storage	Requires more storage space due to data redundancy	More storage-efficient because data is well-organized
Integritas data	Data integrity can be weaker due to duplication	Stronger data integrity as data is more structured
Maintenance (pengisian data dengan proses ETL dari OLTP)	Easier maintenance because of the simpler table structure	More complex maintenance due to multiple related tables

2. Gambar berikut menunjukkan skema OLTP database dari sebuah sistem informasi ekspedisi. Buatlah data warehouse dalam star schema yang digunakan sebagai dasar analisis performa ekspedisi.



Star Schema Design for Expedition Data Warehouse

Fact Table

- FactDelivery → Stores the main transaction data such as delivery and payment.
- Key Columns: DeliveryID (Primary Key)
- Foreign Key that connects to the dimension table:
- SubdistrictOriginID
- DistrictDestinationID
- CourierID
- Delivery StatusID
- PaymentID

Dimension Tables

- DimPayment → Information about the payment.
- DimCourier → Information about the courier.
- DimStatusDelivery → Current status of delivery.
- DimKecamatan → Data of the sub-district area.
- DimKota → City region data.

- DimProvince → Data of the province region.
- DimStatusPayment and DimTypePayment → Details of the status and type of payment.

SQL Implementation

-- Membuat database

```
CREATE DATABASE IF NOT EXISTS ExpeditionWarehouse;
```

```
USE ExpeditionWarehouse;
```

-- Tabel DimStatusPembayaran

```
CREATE TABLE IF NOT EXISTS StatusPembayaran (
  ID INT PRIMARY KEY AUTO_INCREMENT,
  Nama VARCHAR(50) NOT NULL
);
```

-- Tabel DimTipePembayaran

```
CREATE TABLE IF NOT EXISTS TipePembayaran (
  ID INT PRIMARY KEY AUTO_INCREMENT,
  Nama VARCHAR(50) NOT NULL
);
```

-- Tabel DimPembayaran

```
CREATE TABLE IF NOT EXISTS DimPembayaran (
  PembayaranID INT PRIMARY KEY AUTO_INCREMENT,
  Total DECIMAL(10, 2),
  StatusPembayaranID INT,
  TipePembayaranID INT,
  FOREIGN KEY (StatusPembayaranID) REFERENCES StatusPembayaran(ID),
  FOREIGN KEY (TipePembayaranID) REFERENCES TipePembayaran(ID)
);
```

-- Tabel DimKurir

```
CREATE TABLE IF NOT EXISTS Kurir (
  ID INT PRIMARY KEY AUTO_INCREMENT,
  Nama VARCHAR(50) NOT NULL,
  TipeKendaraan VARCHAR(50),
  NoHP VARCHAR(15)
);
```

-- Tabel DimStatusPengiriman

```

CREATE TABLE IF NOT EXISTS StatusPengiriman (
    ID INT PRIMARY KEY AUTO_INCREMENT,
    Nama VARCHAR(50) NOT NULL
);

-- Tabel DimProvinsi
CREATE TABLE IF NOT EXISTS Provinsi (
    ID INT PRIMARY KEY AUTO_INCREMENT,
    Nama VARCHAR(50) NOT NULL
);

-- Tabel DimKota
CREATE TABLE IF NOT EXISTS Kota (
    ID INT PRIMARY KEY AUTO_INCREMENT,
    Nama VARCHAR(50) NOT NULL,
    ProvinsiID INT,
    FOREIGN KEY (ProvinsiID) REFERENCES Provinsi(ID)
);

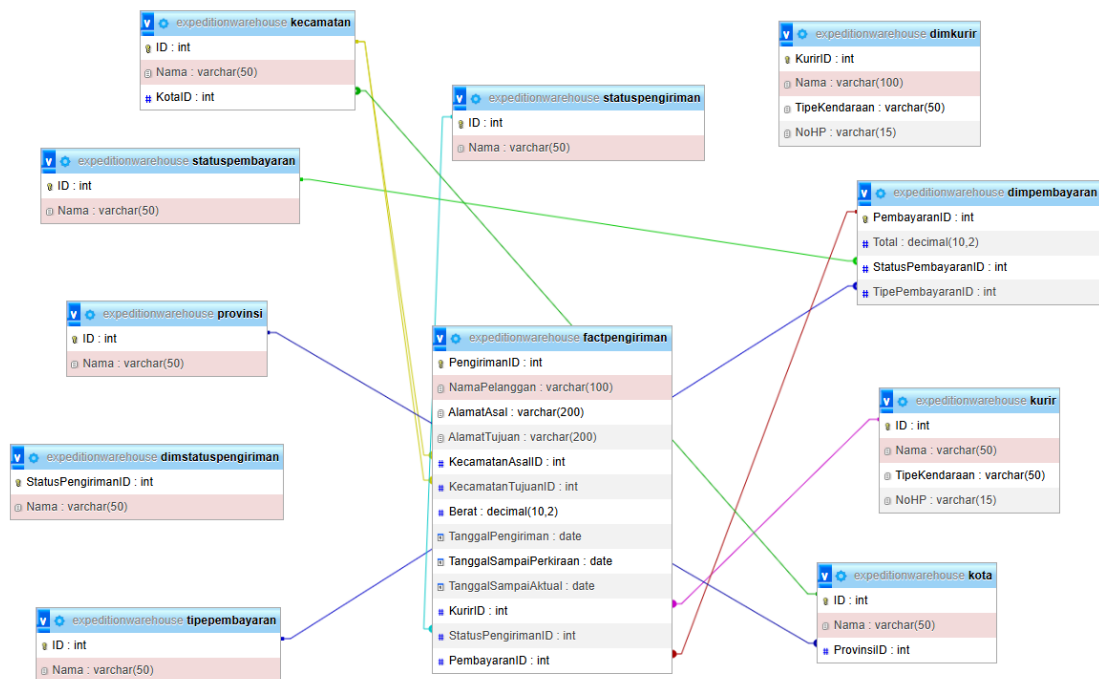
-- Tabel DimKecamatan
CREATE TABLE IF NOT EXISTS Kecamatan (
    ID INT PRIMARY KEY AUTO_INCREMENT,
    Nama VARCHAR(50) NOT NULL,
    KotaID INT,
    FOREIGN KEY (KotaID) REFERENCES Kota(ID)
);

-- Tabel FactPengiriman
CREATE TABLE IF NOT EXISTS FactPengiriman (
    PengirimanID INT PRIMARY KEY AUTO_INCREMENT,
    NamaPelanggan VARCHAR(100) NOT NULL,
    AlamatAsal VARCHAR(200),
    AlamatTujuan VARCHAR(200),
    KecamatanAsalID INT,
    KecamatanTujuanID INT,
    Berat DECIMAL(10, 2),
    TanggalPengiriman DATE,
    TanggalSampaiPerkiraan DATE,
    TanggalSampaiAktual DATE,

```

KurirID INT,
 StatusPengirimanID INT,
 PembayaranID INT,
 FOREIGN KEY (KecamatanAsalID) REFERENCES Kecamatan(ID),
 FOREIGN KEY (KecamatanTujuanID) REFERENCES Kecamatan(ID),
 FOREIGN KEY (KurirID) REFERENCES Kurir(ID),
 FOREIGN KEY (StatusPengirimanID) REFERENCES StatusPengiriman(ID),
 FOREIGN KEY (PembayaranID) REFERENCES DimPembayaran(PembayaranID)
);

Entity Relationship Diagram (ERD)



The reason why the DimStatusDelivery and DimCourier tables do not have direct relationships to other tables outside FactDelivery is because both tables function as standalone dimension tables that only provide additional information related to delivery transactions.

Implementation of DimKurir Table in Data Warehouse with Pentaho

Inserting the DimKurir table in phpMyAdmin

		KurirID	Nama	TipeKendaraan	NoHP
☐	Edit Copy Delete	201	Nayeon	Motorcycle	81234567890
☐	Edit Copy Delete	202	Jeongyeon	Car	81987654321
☐	Edit Copy Delete	203	Momo	Truck	81123456789

export a CSV file named dimkurir.csv

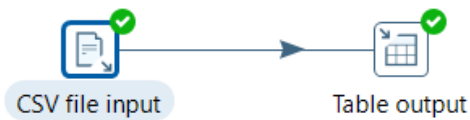
Pentaho Data Integration (PDI Spoon) Configuration

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim ty
1	KurirID	Integer	#	15	0	\$	.	,	none
2	Nama	String		9		\$	.	,	none
3	TipeKendaraan	String		10		\$	.	,	none
4	NoHP	Integer	#	15	0	\$	.	,	none

To process data using the CSV File Input component, open the settings by double-clicking it, then on the File tab, select a CSV file such as dimkurir.csv, check Header row present so that the first row is read as a header, then on the Fields tab, click Get Fields to automatically detect columns, and end by clicking OK to save the settings.

To configure the Table Output component, double-click it to open its settings. In the Connection section, click New, then fill in the parameters: Connection Name "Connection_Expedition", Connection Type MySQL, Host Name localhost, Database Name ExpeditionWarehouse, Port 3306, User Name root, and leave the Password blank if there is none. Click Test to ensure a successful connection, then click OK. Select the DimKuririr table in the Target Table, go to the Database Fields tab, click Get Fields to customize the columns with data from the CSV, and click OK to save.

Executing Transformation



Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2025/03/25 02:16:57 - Spoon - Transformation opened.
2025/03/25 02:16:57 - Spoon - Launching transformation [Transformation 1]...
2025/03/25 02:16:57 - Spoon - Started the transformation execution.
2025/03/25 02:16:57 - Transformation 1 - Dispatching started for transformation [Transformation 1]
2025/03/25 02:16:57 - Table output.0 - Connected to database [quiz1] (commit=1000)
2025/03/25 02:16:57 - CSV file input.0 - Header row skipped in file 'C:\SEMESTER 4\dimkurir (1).csv'
2025/03/25 02:16:57 - CSV file input.0 - Finished processing (I=4, O=0, R=0, W=3, U=0, E=0)
2025/03/25 02:16:57 - Table output.0 - Finished processing (I=0, O=3, R=3, W=3, U=0, E=0)
2025/03/25 02:16:57 - Spoon - The transformation has finished!!

The result

#	KurirID	Nama	TipeKendaraan	NoHP
1	201	Nayeon	Motorcycle	81234567890
2	202	Jeongyeon	Car	81987654321
3	203	Momo	Truck	81123456789

				KurirID	Nama	TipeKendaraan	NoHP
☐	Edit	Copy	Delete	201	Nayeon	Motorcycle	81234567890
☐	Edit	Copy	Delete	202	Jeongyeon	Car	81987654321
☐	Edit	Copy	Delete	203	Momo	Truck	81123456789